

6 A transformation T of the plane has associated matrix $\mathbf{M} = \begin{pmatrix} 1 & \lambda + 1 \\ \lambda - 1 & -1 \end{pmatrix}$, where λ is a non-zero constant.

(a) (i) Show that T reverses orientation. [3]

(ii) State, in terms of λ , the area scale factor of T . [1]

(b) (i) Show that $\mathbf{M}^2 - \lambda^2 \mathbf{I} = \mathbf{0}$. [2]

(ii) Hence specify the transformation equivalent to two applications of T . [1]

(c) In the case where $\lambda = 1$, T is equivalent to a transformation S followed by a reflection in the x -axis.

(i) Determine the matrix associated with S . [3]

(ii) Hence describe the transformation S . [2]